

Virtual Microbial Culture Research on Escherichia coli

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Abstract

The *Escherichia coli* bacterium is among the widely researched bacteria in both biotechnology and microbiology (NCBI, 2023). While most of them are non-harmful bacteria that are part of the normal intestinal flora of mammals, some of the pathogenic types of *E. coli* may cause gastroenteritis and urinary tract infections. Because of its fast reproduction rate, nutritional simplicity, and known genetics, *E. coli* has been a major model organism for both biotechnology and genetic engineering. This virtual microbial culture lab looks at the growth of *Escherichia coli* bacteria (Keseler et al., 2021).

Introduction

Escherichia coli is a Gram-negative, rod-shaped bacterium that belongs to the family of Enterobacteriaceae (NCBI, 2023). The natural habitat of *E. coli* is the intestines of humans and warm-blooded animals. The majority of *E. coli* strains play an important role in the maintenance of intestinal flora. But certain *E. coli* strains like Enterohemorrhagic *Escherichia coli* (EHEC), Enterotoxigenic *Escherichia coli* (ETEC) and Uropathogenic *Escherichia coli* (UPEC) are able to cause dangerous diseases (NCBI, 2023).

In biotechnology *E. coli* is the organism that is widely used as one of the host organisms for recombinant DNA technology (Keseler et al., 2021). Genetically engineered *E. coli* is often employed to produce insulin, growth hormone, enzymes, vaccines and other recombinant proteins due to rapid growth of *E. coli*, inexpensive growth medium and detailed studies of its genome (Keseler et al., 2021).

The purpose of this virtual experiment is the study of cultivation features of *E. coli* under laboratory conditions and morphological and biotechnological applications of *E. coli*.

Objectives

The aims of this virtual microbial culture lab exercise include:

- Studying the growth patterns of *E. coli*.
- Colony morphology in nutrient agar.
- Bacterial cell morphology by Gram staining.
- Significance of *E. coli* in biotechnology and microbiology.
- Laboratory observations without the presence of live microorganisms.

Materials

- Nutrient agar plate
- MacConkey agar plate
- Sterile inoculating loop
- Incubator (37°C)
- Microscope
- Gram staining reagents
- Glass slides
- Immersion oil
- Personal protective equipment (PPE)

Methodology

STEP 1: PREPARATION OF CULTURE MEDIA

Nutrient agar and MacConkey agar plates were prepared based on the standard procedure in microbiology. The nutrient agar and MacConkey agar plates will be used as media which contain nutrients needed by bacteria. MacConkey agar is used to differentiate lactose fermenters.

STEP 2: INOCULATION

A pure culture of *E. coli* was inoculated into both plates using the quadrant streaking method.

STEP 3: INCUBATION

Inoculated plates were then incubated at 37° Celsius for 24 hours under aerobic conditions.

STEP 4: OBSERVATION OF COLONIES

After incubation, the following characteristics were observed regarding colonies:

- Size
- Shape
- Color
- Margin
- Surface elevation
- Surface texture

Lactose fermentation on MacConkey agar was also checked.

STEP 5: GRAM STAINING

Gram staining of the bacterial smear was carried out using:

1. Crystal violet
2. Iodine solution
3. Decolorization with alcohol
4. Safranin counterstaining

Stained bacteria were then examined using oil immersion (1000x magnification).

Results

COLONY MORPHOLOGY ON NUTRIENT AGAR

E. coli colonies were characterized after 24 hours of incubation by:

Characteristic	Observation
Shape	Circular
Size	2-4 mm diameter
Color	Cream to off-white
Texture	Moist and shiny
Surface	Smooth
Height	Slightly convex

Growth on MacConkey Agar

There was growth of *E. coli* on MacConkey Agar with pink colonies, showing that there is positive lactose fermentation.

This result proves one of the biochemical characteristics of *E. coli*.

Microscopic Observation

AFTER GRAM STAINING, BACTERIA SHOWED:

- Gram -Negative
- Pink Color
- Rod Shaped (Bacilli)
- Single and with few chains

Discussion

The experimental data received in the virtual lab are compatible with the microbiological properties of *Escherichia coli* (NCBI, 2023).

Smooth, round colonies growing in nutrient agar are common for good cultures of *E. coli*. Pink colonies in MacConkey agar indicate lactose fermentation that helps to distinguish *E. coli* from many other Gram-negative bacteria (NCBI, 2023).

Under the microscope after Gram staining, Gram-negative bacilli were seen, which is connected with the thin cell wall and presence of an additional outer membrane of *E. coli* (NCBI, 2023).

However, apart from microbiological properties, *Escherichia coli* is extremely valuable in biotechnology. The bacterium is widely used as a host cell in recombinant DNA technology due to the possibility of introducing plasmids to *E. coli* (Keseler et al., 2021).

SOME OF THE USEFUL APPLICATIONS OF BIOTECHNOLOGY ARE:

- Recombinant human insulin production
- Human growth hormone production
- Antigen production for vaccines
- Enzyme production
- Gene cloning
- CRISPR

- DNA sequencing
- Synthetic biology

E. coli is one of the most useful model organisms in biological research because it has been studied genetically to an extent that is unprecedented (Keseler et al., 2021).

Advantages of Using *E. coli* in Biotechnology

- Fast growth (growth time is approximately 20 minutes under optimum conditions)
- Nutrient needs are simple
- Genetic manipulation is easy
- Cultivation is inexpensive
- High protein synthesis
- Wide range of scientific information is available
- Safe laboratory strains available (e.g., *E. coli* K-12) (Keseler et al., 2021).

Limitations

However, there are some drawbacks to *E. coli* too:

- Certain proteins cannot be folded correctly.
- The level of post-translational modifications is limited.
- Endotoxins need purification during drug manufacture.
- Pathogenic strains require a higher biosafety level (NCBI, 2023).

Conclusion

In the aforementioned experiment, the growth properties of *E. coli* have been clearly illustrated. They produced smooth cream-colored colonies on the nutrient agar media, reduced lactose on MacConkey agar media, and had a rod shape on the Gram-staining technique (NCBI, 2023).

E. coli is one of the organisms that have immense value in the field of biotechnology, which was shown during the experiment. This is due to the fast-growing nature of the organism, its genetics, and genetic modification. All these features make *E. coli* useful in recombinant protein production, vaccines, molecular biology, and genetic engineering despite the presence of pathogenic strains (Keseler et al., 2021).

References

National Center for Biotechnology Information. (2023). *Escherichia coli*. In StatPearls. StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK562250/>

Keseler, I. M., Gama-Castro, S., Mackie, A., et al. (2021). The EcoCyc database in 2021. *Nucleic Acids Research*, 49(D1), D605–D612. <https://doi.org/10.1093/nar/gkaa1004>